Hukum Ampere



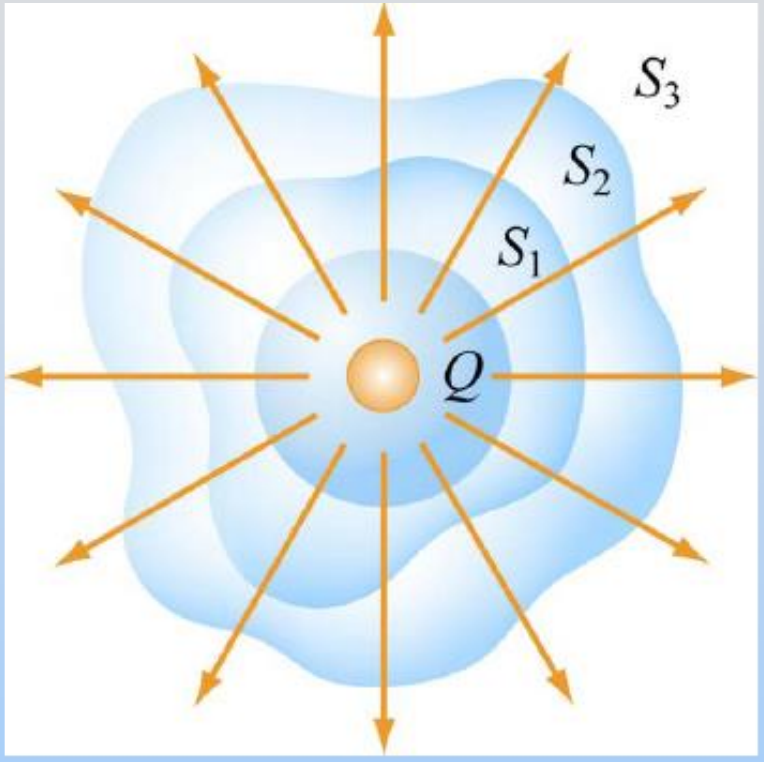
Dosen : Defrianto Pratama, S.Pd, M.Si.

Teknik Elektronika

Politeknik Negeri Bandung

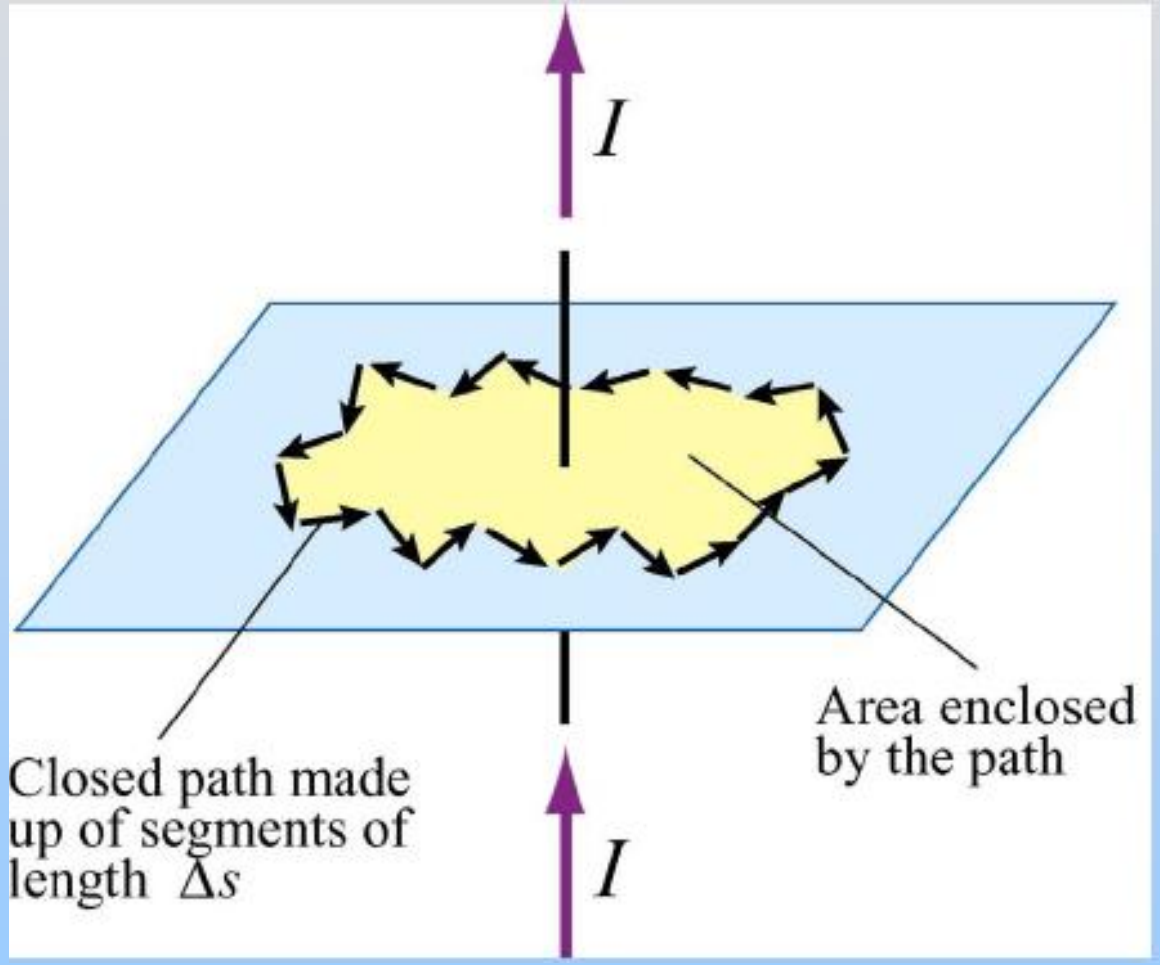
Sumber bacaan Ebook: Halliday ch 29

Review :
Hukum Gauss – Idenya



Total “flux” yang menembus setiap permukaan adalah sama dan hanya bergantung pada jumlah muatan di dalam permukaan tersebut

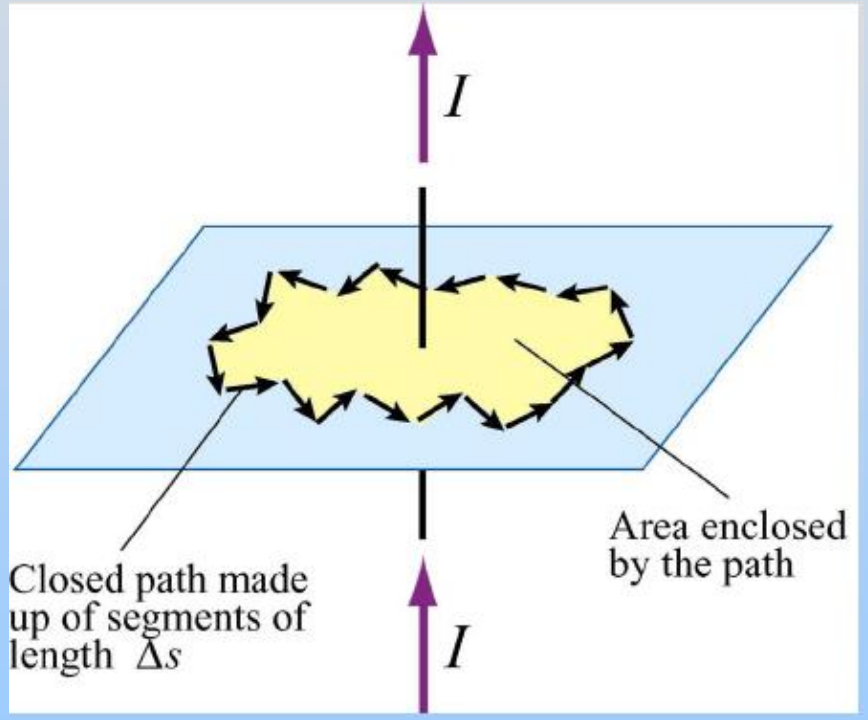
Hukum Ampere: Idenya



Untuk mencari medan B disekeliling loop, pastilah terdapat arus yang menembus loop

Hukum Ampere: Persamaan

$$\oint \vec{\mathbf{B}} \cdot d\vec{\mathbf{s}} = \mu_0 I_{enc}$$



Integral garisnya sepanjang lintasan tertutup yang mengelilingi sebuah permukaan terbuka *S.*

I_{enc} adalah arus yang menembus S:

$$I_{enc} = \int_S \vec{\mathbf{J}} \cdot d\vec{\mathbf{A}}$$

Biot-Savart vs. Ampere

Hukum Biot-Savart

$$\vec{\mathbf{B}} = \frac{\mu_0 I}{4\pi} \int \frac{d\vec{\mathbf{s}} \times \hat{\mathbf{r}}}{r^2}$$

Sumber arus umum (sembarang)
contoh: kawat berhingga, kawat loop

Hukum Ampere

$$\oint \vec{\mathbf{B}} \cdot d\vec{\mathbf{s}} = \mu_0 I_{enc}$$

Sumber arus simetri
contoh: kawat takberhingga, lembaran arus takhingga

Penggunaan Hukum Ampere

1. Identifikasi daerah yang akan dihitung medan B nya

   Mencari arah B dengan aturan tangan kanan

2. Pilih lintasan Ampere S: Simetri

   B = 0 atau konstan pada lintasan Ampere!

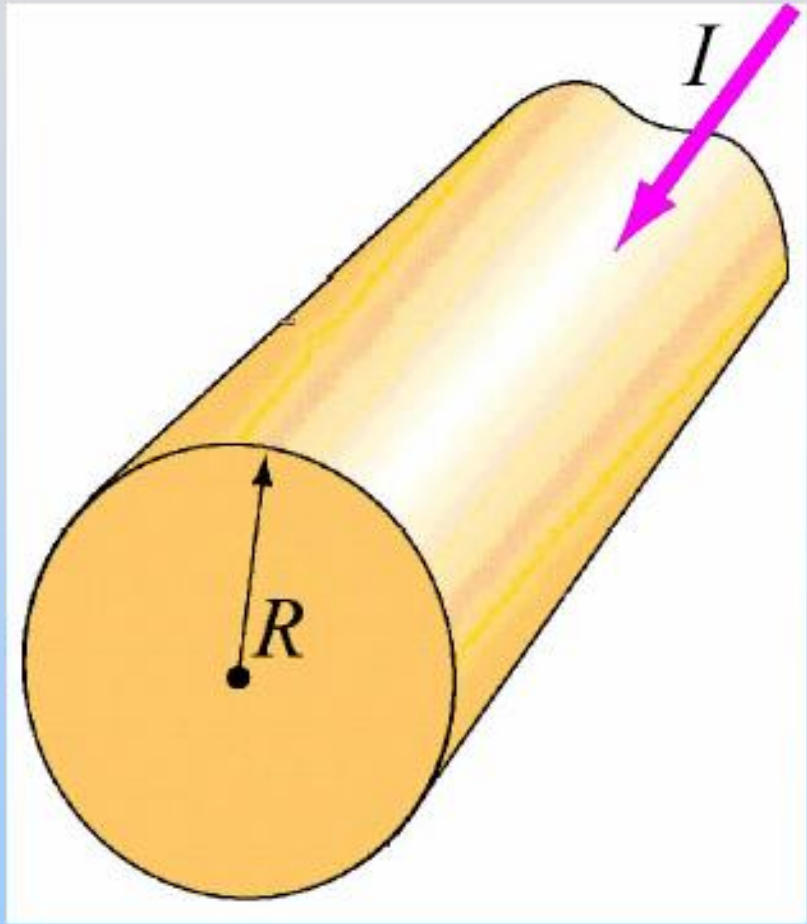
3. Hitung $\oint \vec{\mathbf{B}} \cdot d\vec{\mathbf{s}}$

4. Hitung arus yang dilingkupi lintasan Ampere S

5. Gunakan hukum Ampere untuk mencari B

$$\oint \vec{\mathbf{B}} \cdot d\vec{\mathbf{s}} = \mu_0 I_{enc}$$

Contoh: Kawat tak Berhingga



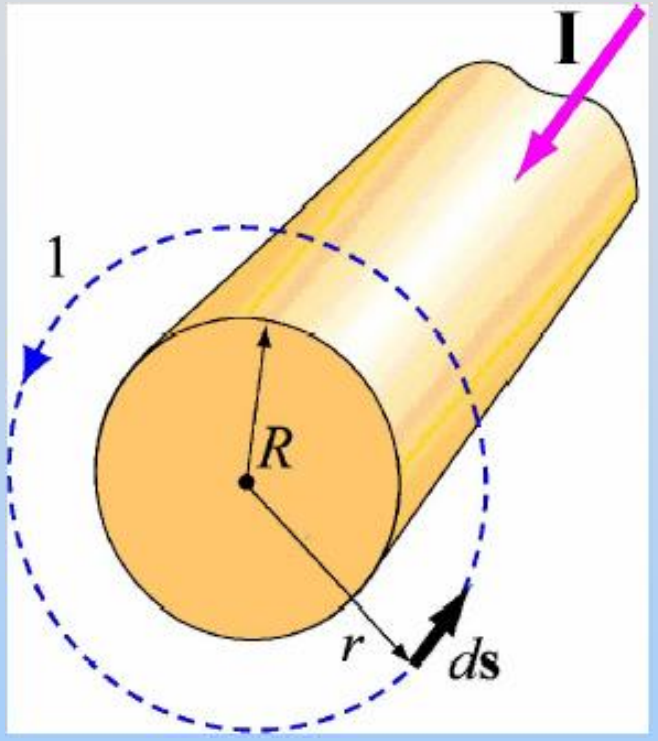
Sebuah konduktor silinder memiliki radius R dan rapat arus uniform dengan arus total I

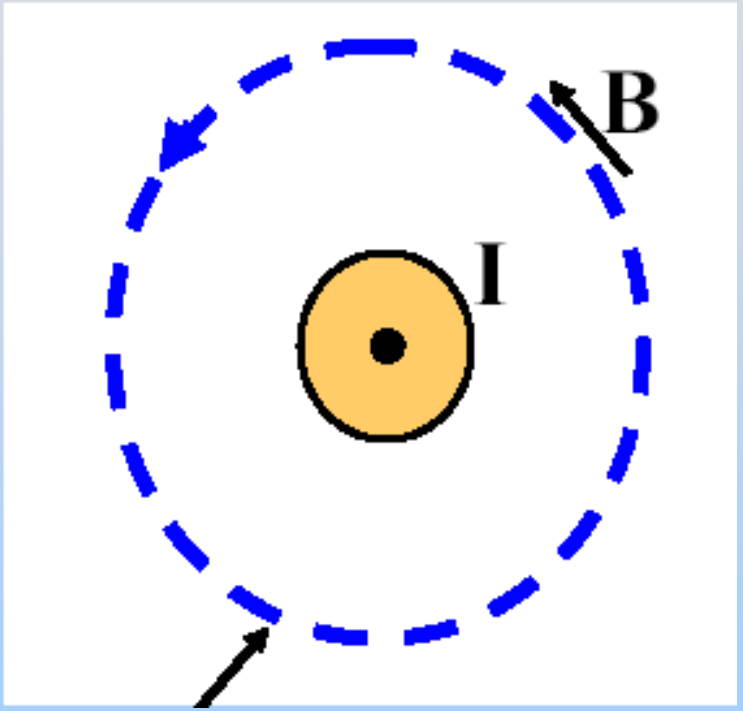
Cari B di daerah:

(1) Luar kawat ($r \geq R$)

(2) Dalam kawat ($r < R$)

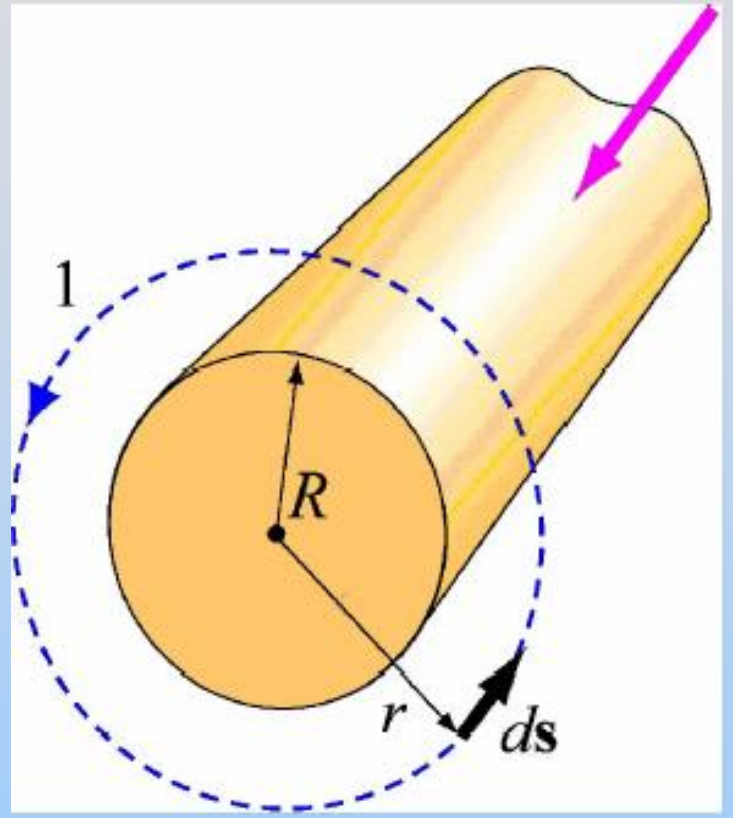
Contoh Hukum Ampere: Kawat Takhingga





Lintasan Ampere:
B konstan & sejajar
I terlingkupi

Contoh: Kawat Radius *R*



Daerah 1: $r \geq R$

Simetri Silinder $\rightarrow$

Lintasan Ampere lingkaran

Medan B berlawanan jarum jam

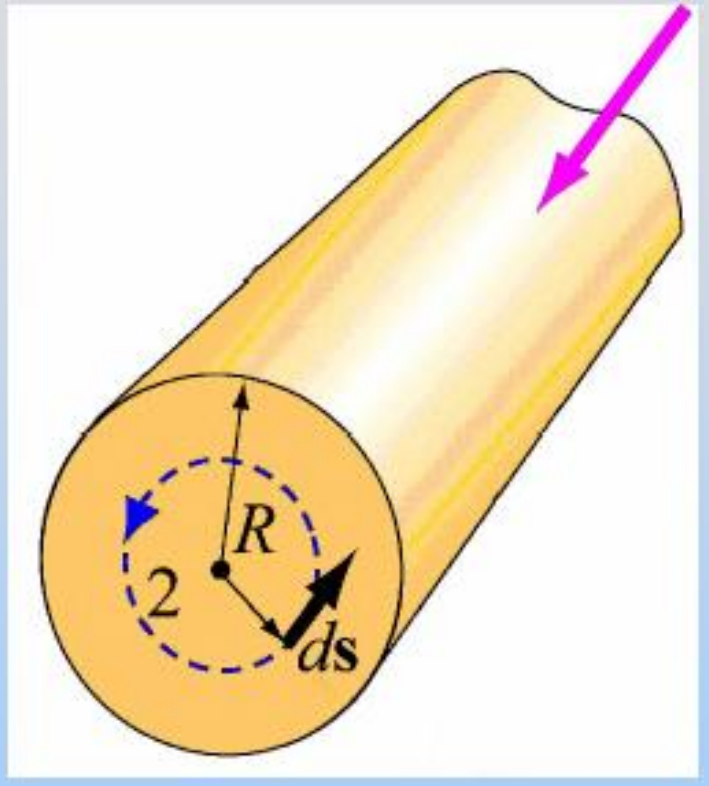
$$\oint \vec{\mathbf{B}} \cdot d\vec{\mathbf{s}} = B \oint d\vec{\mathbf{s}} = B(2\pi r)$$

$$= \mu_0 I_{enc} = \mu_0 I$$

$$\vec{\mathbf{B}} = \frac{\mu_0 I}{2\pi r}$$

Berlawanan arah jarum jam

Contoh: Kawat Radius *R*



Daerah 2: r < R

$$\oint \vec{\mathbf{B}} \cdot d\vec{\mathbf{s}} = B \oint d\vec{\mathbf{s}} = B(2\pi r)$$

$$= \mu_0 I_{enc} = \mu_0 I \left(\frac{\pi r^2}{\pi R^2} \right)$$

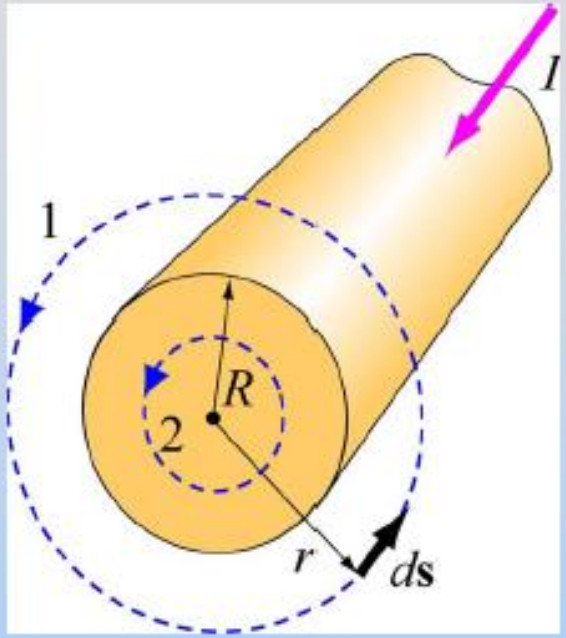
$$\vec{\mathbf{B}} = \frac{\mu_0 I r}{2\pi R^2}$$

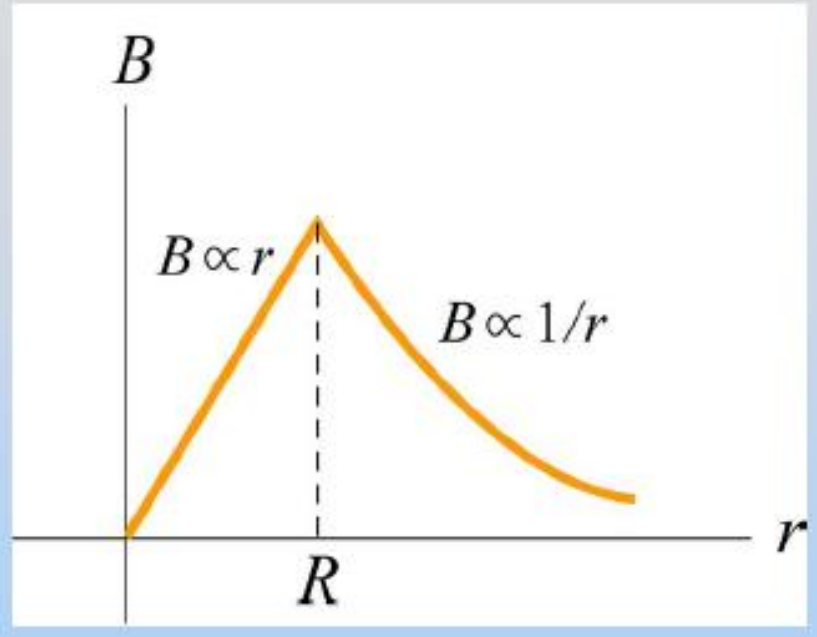
Berlawanan arah jarum jam

Dapat juga:

$$J = \frac{I}{A} = \frac{I}{\pi R^2}; \; I_{enc} = JA_{enc} = \frac{I}{\pi R^2}\left(\pi r^2\right)$$

Contoh: Kawat Radius *R*

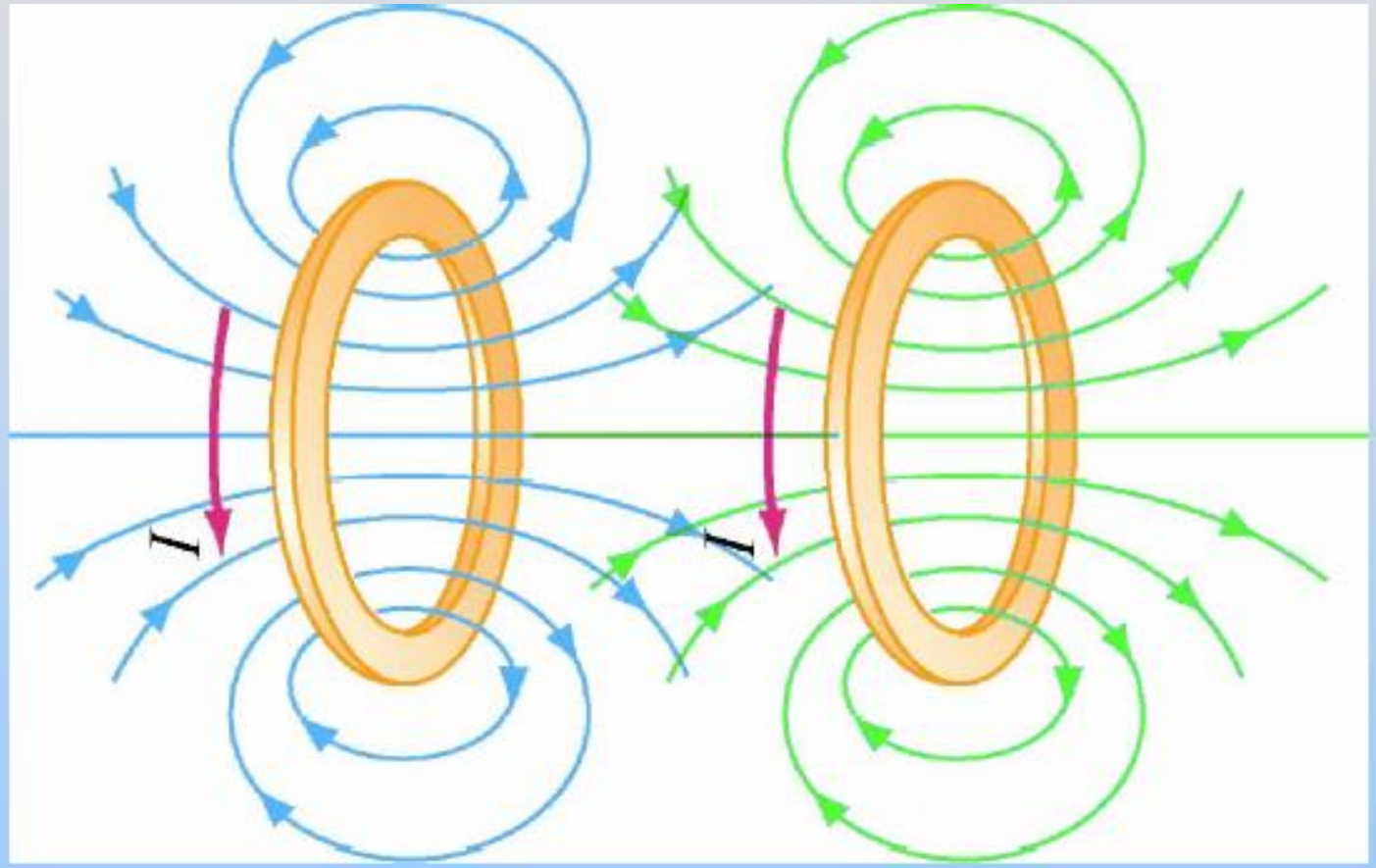




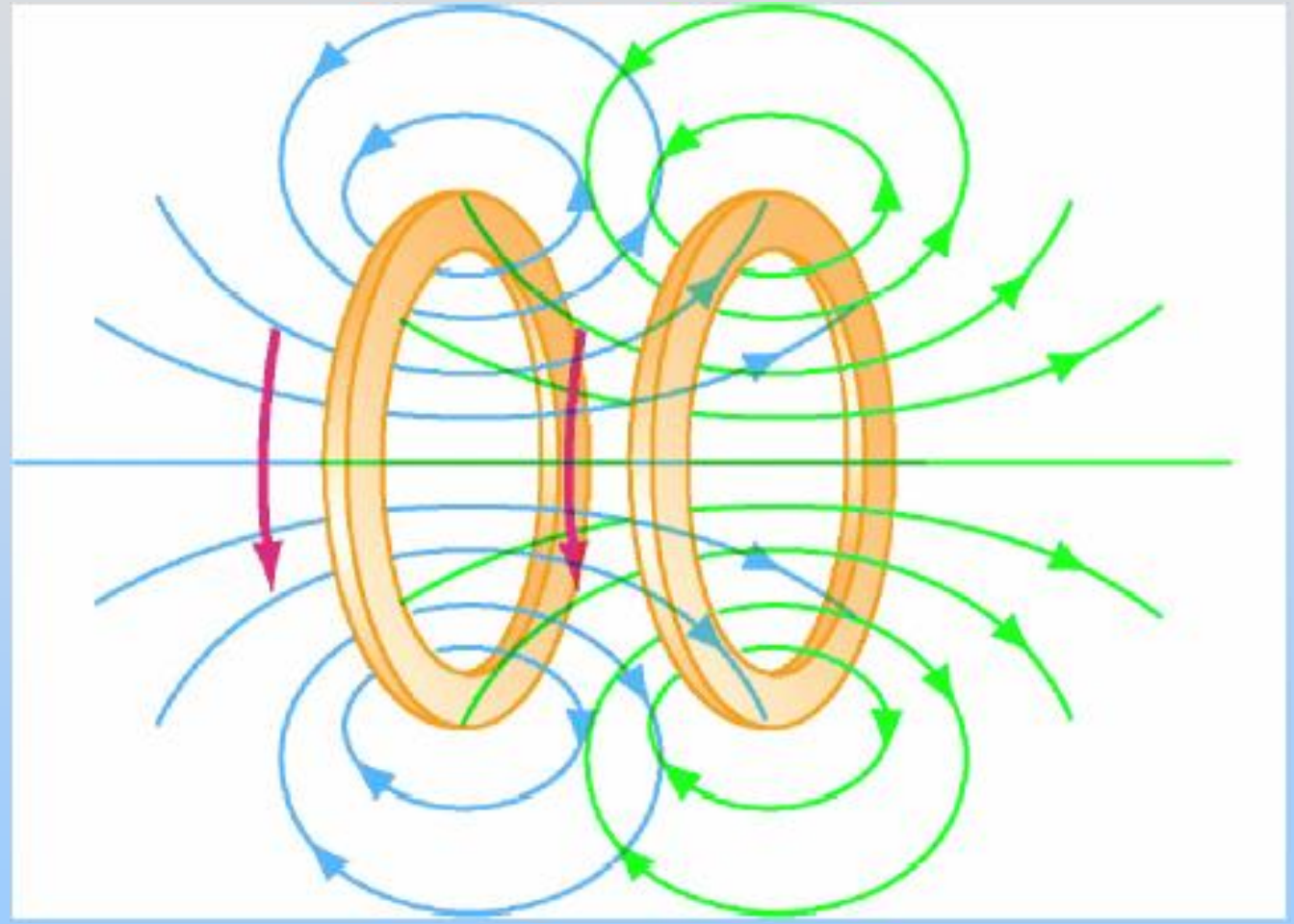
$$B_{in} = \frac{\mu_0 I r}{2\pi R^2} \qquad B_{out} = \frac{\mu_0 I}{2\pi r}$$

Geometri yang Lain

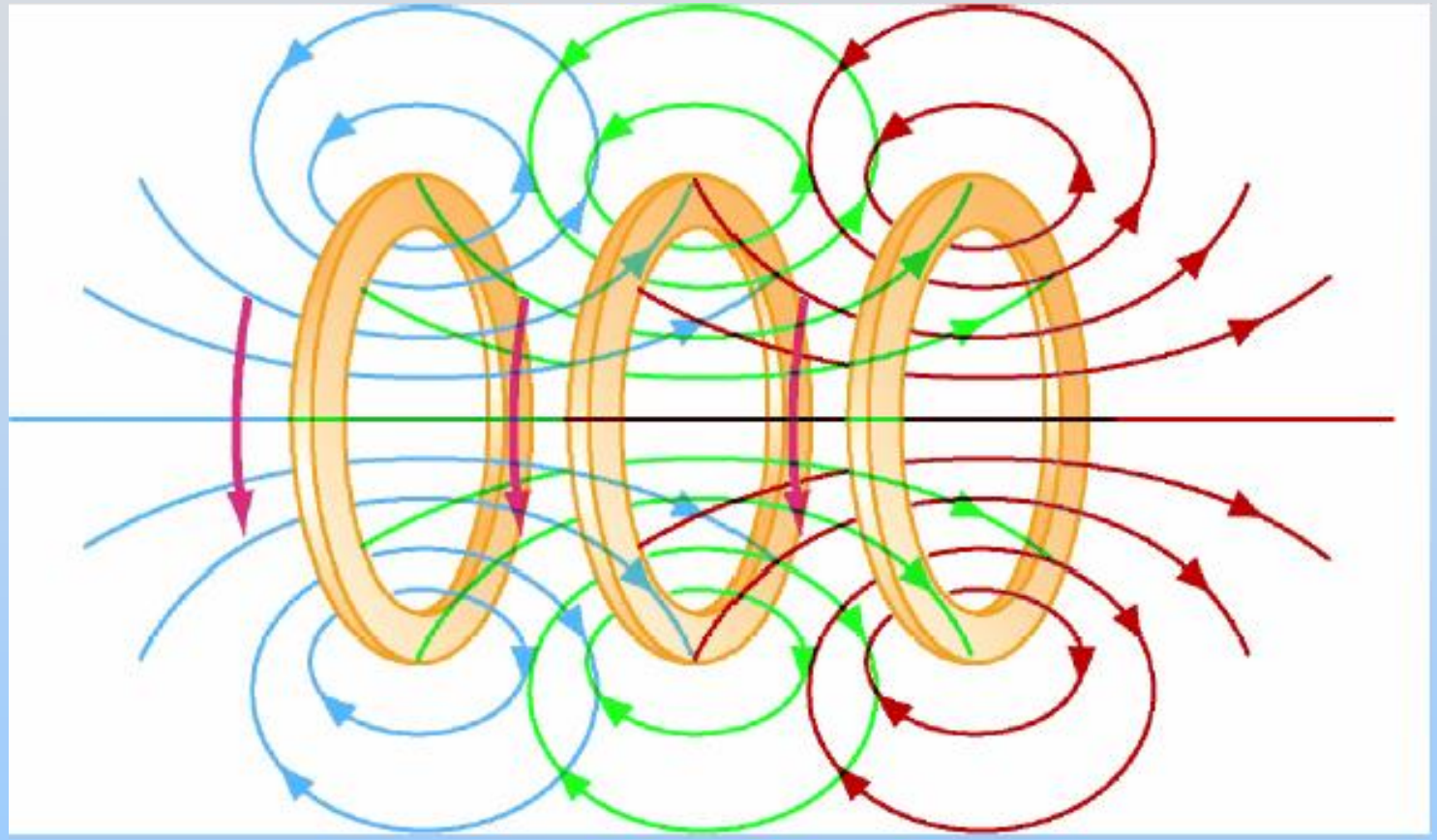
Koil Helmholtz



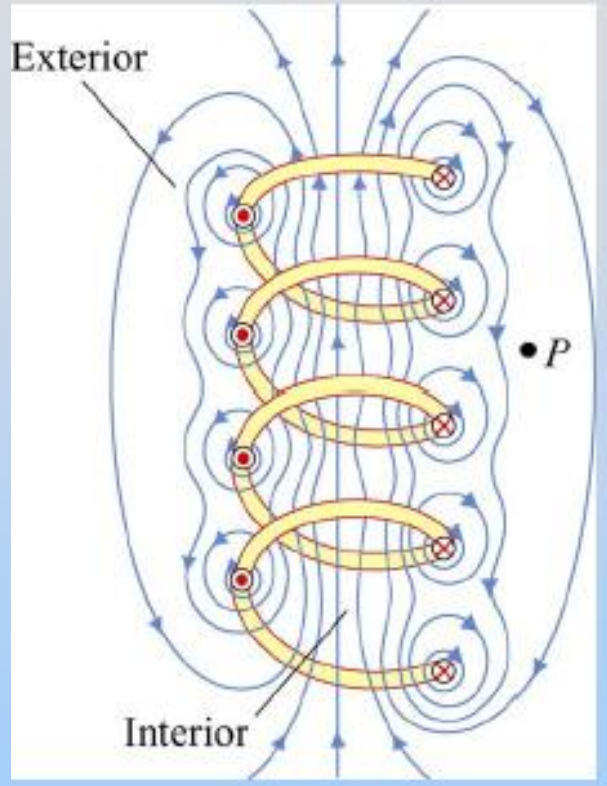
Koil Helmholtz dengan Jarak Lebih Dekat

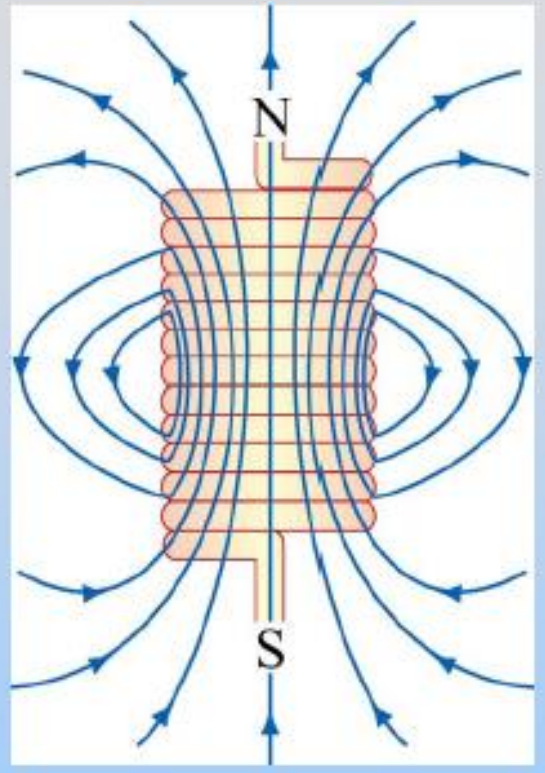


Lilitan Kawat



Medan Magnet dari Solenoid

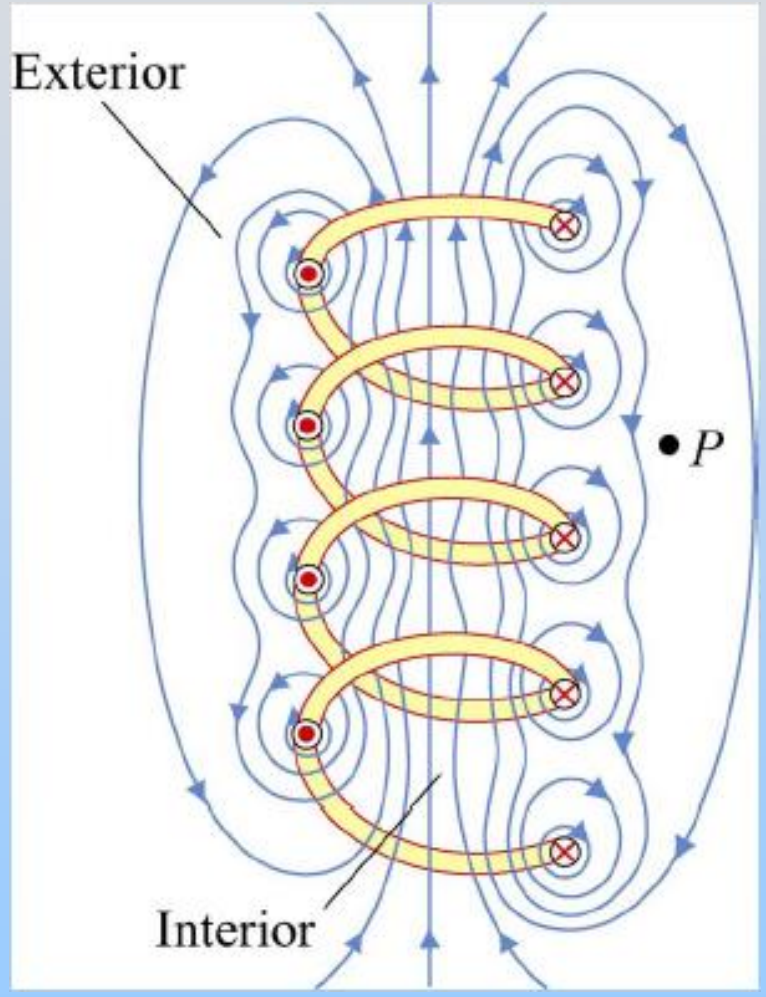




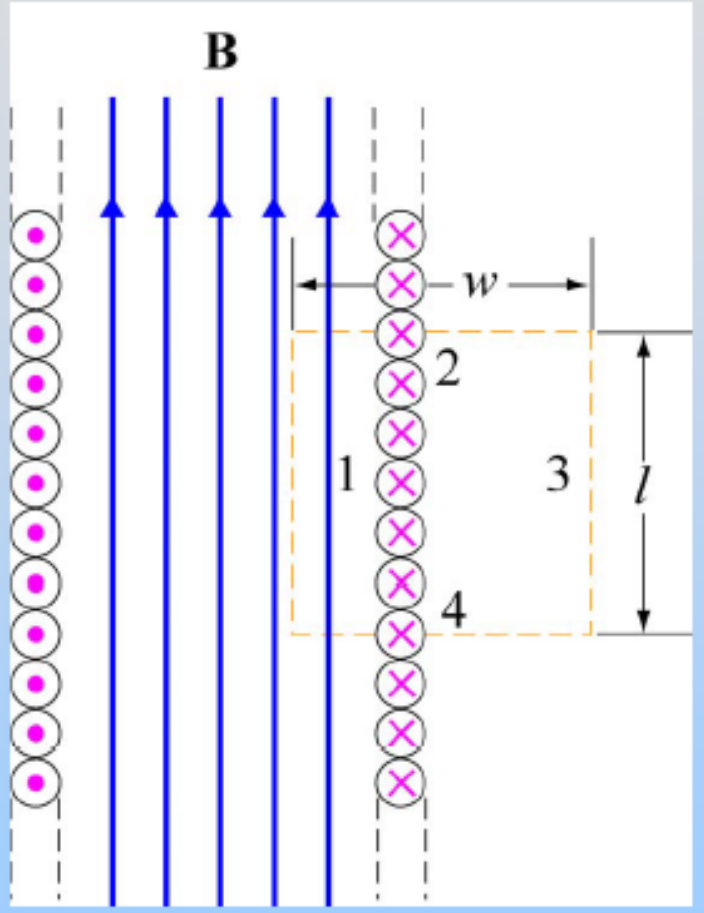
Gulungan renggang Gulungan rapat

Untuk solenoid ideal, B uniform di dalam & nol di luar

Lilitan Kawat – Solenoid



Medan Magnet dari Solenoid



Gunakan hukum Ampere:

$$\begin{cases} \vec{\mathbf{B}} \perp d\vec{\mathbf{s}} & \text{along sides 2 and 4} \\ \vec{\mathbf{B}} = 0 & \text{along side 3} \end{cases}$$

$$\oint \vec{\mathbf{B}} \cdot d\vec{\mathbf{s}} = \int_1 \vec{\mathbf{B}} \cdot d\vec{\mathbf{s}} + \int_2 \vec{\mathbf{B}} \cdot d\vec{\mathbf{s}} + \int_3 \vec{\mathbf{B}} \cdot d\vec{\mathbf{s}} + \int_4 \vec{\mathbf{B}} \cdot d\vec{\mathbf{s}}$$

$$= \quad Bl \quad + \quad 0 \quad + \quad 0 \quad + \quad 0$$

$$I_{enc} = nlI$$

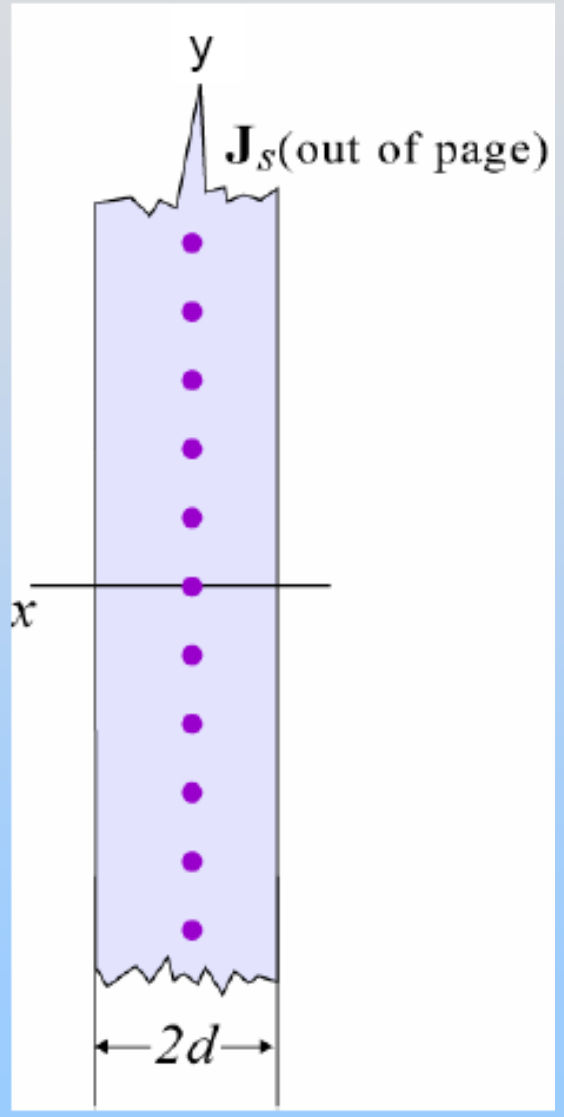
n: rapat lilitan

$n = N/L$

$$\oint \vec{\mathbf{B}} \cdot d\vec{\mathbf{s}} = Bl = \mu_0 nlI$$

$$\boxed{B = \frac{\mu_0 nlI}{l} = \mu_0 nI}$$

Problem: Lembaran Arus

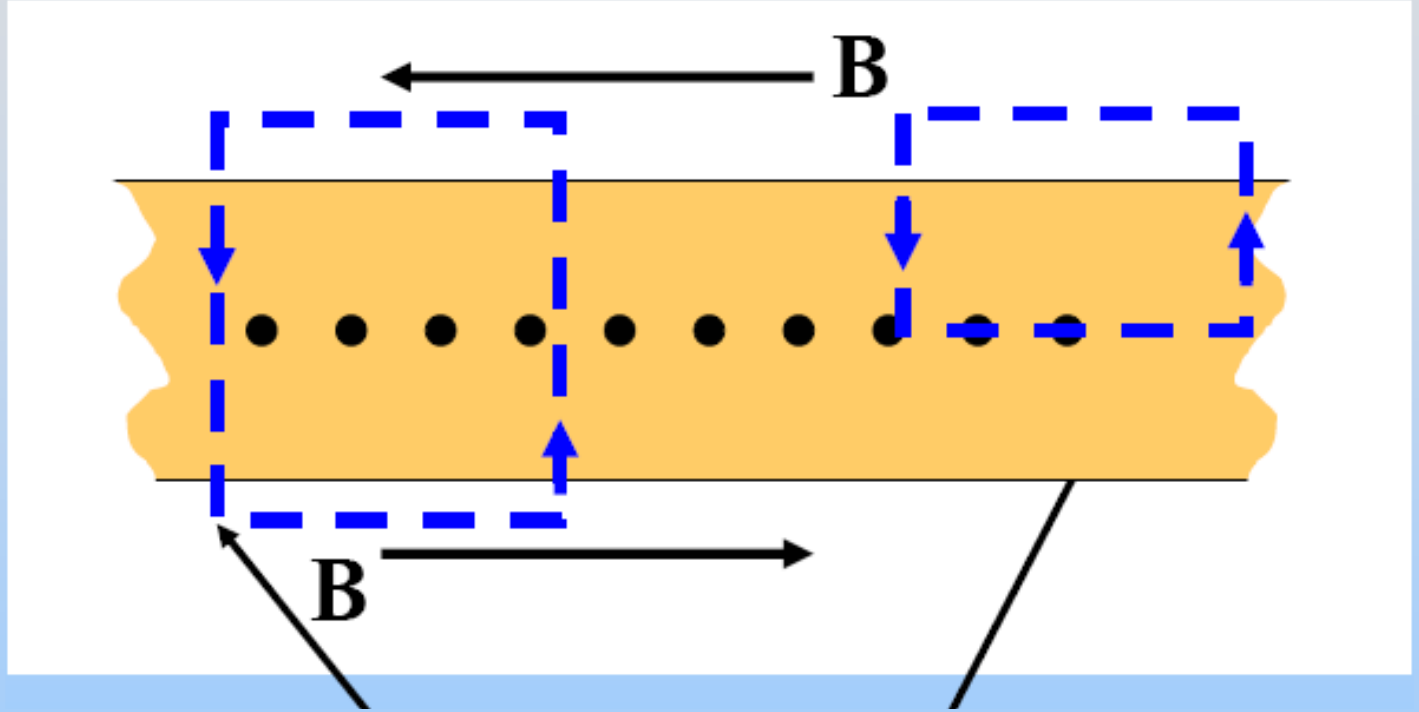


Sebuah lembaran arus (arah y & z takhingga, tebal 2d arah x) membawa rapat arus uniform:

$$\vec{\mathbf{J}}_s = J\hat{\mathbf{k}}$$

Cari B di luar dan di dalam!

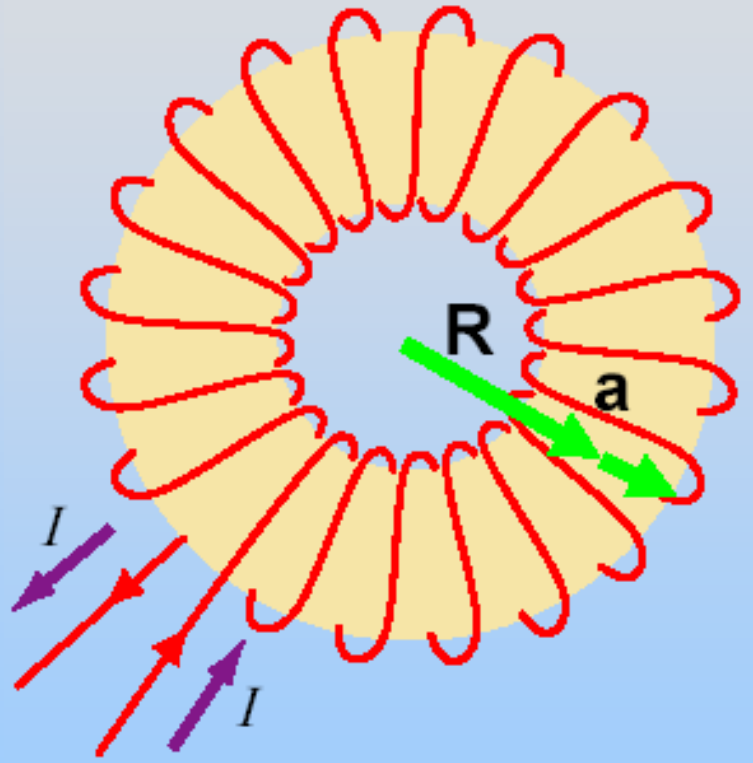
Hukum Ampere:
Lembaran Arus Takhingga



Lintasan Ampere:

B Konstan & Paralel OR Tegaklurus OR Nol
Arus Terlingkupi

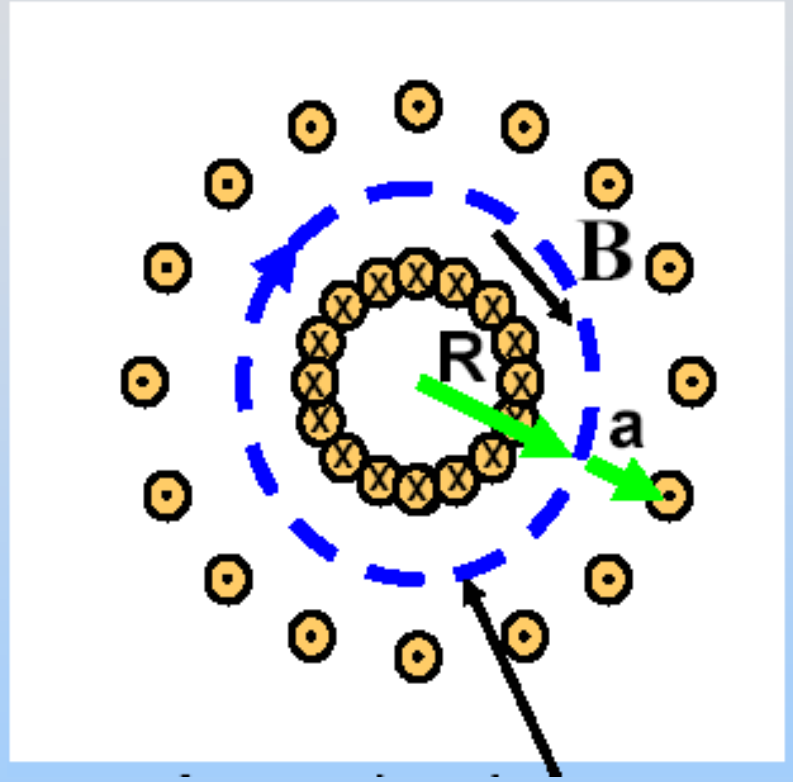
Problem: Torus



Sebuah torus (sebuah solenoida radius a dan n lilitan/meter yang ujung-ujungnya diketemukan sehingga membentuk donat berradius R) berarus uniform I.

Cari B disumbu pusat solenoid!

Hukum Ampere: Torus

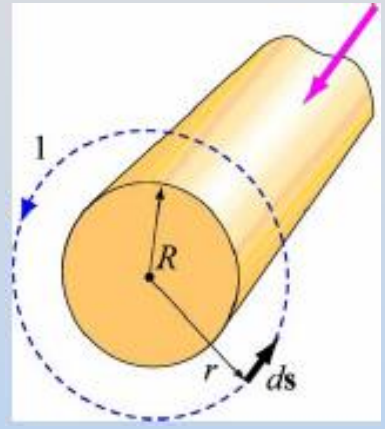


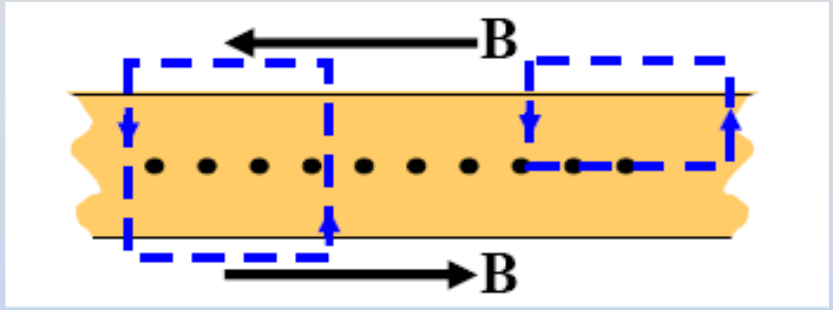
Lintasan Ampere:
B konstan & Paralel
I terlingkupi

Hukum Ampere:

$$\oint \vec{\mathbf{B}} \cdot d\vec{\mathbf{s}} = \mu_0 I_{enc}$$

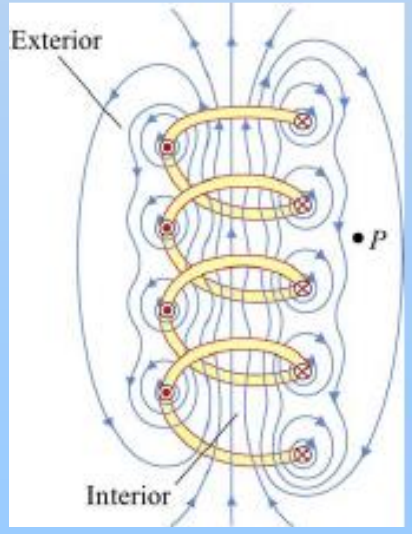
Kawat Silinder Takhingga

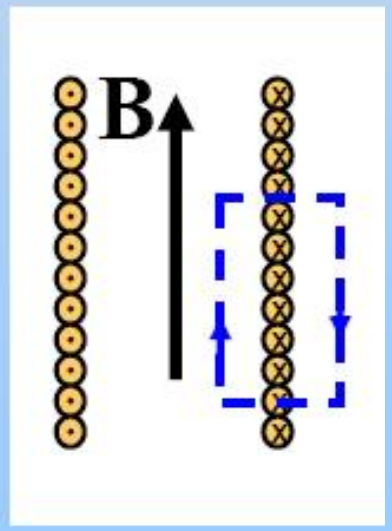


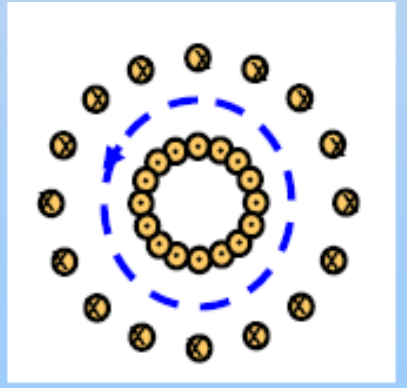


Lembaran Arus Takhingga

Solenoid

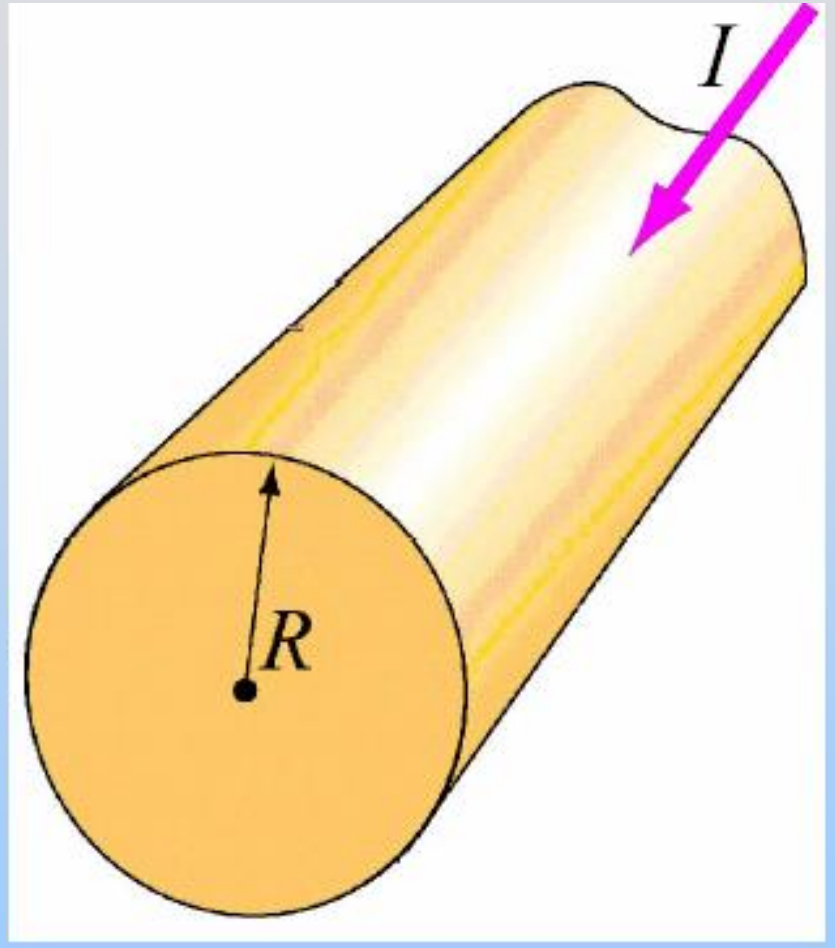






Torus

TUGAS



Sebuah kawat silinder radius R dan rapat arus nonuniform dengan arus total:

$$\vec{\mathbf{J}} = J_0 \frac{R}{r}$$

Cari B di daerah:
(1) r ≥ R
(2) r < R